

Comb-pumped backward Brillouin storage of frequency-bin photons in thin-film lithium niobate: delay figures of merit and a phase-matching limit on mode selectivity

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We analyze, using simulations and closed-form estimates only, a backward stimulated-Brillouin-scattering (SBS) memory in thin-film lithium niobate (TFLN) pumped by an electro-optic frequency comb, proposed as an all-optical, transducer-free buffer for frequency-bin photons. With measured TFLN parameters the acoustic delay loss at 4 K is 2.2 dB/ μ s, which is 78 $\times$ and 12 $\times$ below record and material-limited TFLN optical delay lines (169 and 26 dB/ μ s), and 1 μ s of acoustic delay occupies 3.1 mm. Off-chip fiber is still 53 $\times$ better in loss (0.04 dB/ μ s) but requires 204 m. A full photon–phonon swap needs 16 mW of peak pump for 10-MHz photons and 1.55 W for 100-MHz photons, which restricts the scheme to narrowband photons. A thermal-noise model excludes single-photon operation at room temperature and gives a noise-limited qubit fidelity above 0.9 for storage up to 40 ns at 4 K. At 20 mK the fidelity stays above 0.99 up to 10 μ s, under the untested assumption that the 4-K acoustic loss carries over to millikelvin temperatures. A single-acoustic-mode model predicts that the comb-line amplitudes and phases select which frequency-bin superposition is stored, with selectivity 0.9988 for eight bins. We show, however, that this local model fails for backward SBS. Bins spaced by Δ write phonon gratings whose wave vectors differ by $2\Delta n_g/c$ (964 rad/m for 10-GHz bins), and these gratings are mutually orthogonal over the meter-scale collision region that nanosecond photons require. A space–time simulation confirms that the two-bin selectivity collapses to 0.50. What survives is a bin-diagonal memory with per-bin programmable swap, phase, and read time. It supports re-timing and synchronization of photons across bins, but it cannot remap frequency or select superpositions.

I. INTRODUCTION

Frequency-bin encoding places many qubit or qudit modes in a single waveguide mode and is naturally phase stable [1]. Programmable frequency-bin gates have been demonstrated with electro-optic (EO) quantum frequency processors [2, 3], EO frequency beam splitters on TFLN [4, 5], and temporal-mode-selective sum-frequency generation in TFLN quantum pulse gates [6]. None of these devices can store a photon. On-chip TFLN storage has so far relied on optical recirculation [7] or long delay lines [8], whose loss per unit delay is set by the optical propagation loss [9, 10].

Brillouin light storage avoids this limit by transferring the optical signal to a traveling acoustic wave that moves $\sim 10^5$ times more slowly than light. It was demonstrated in fiber [11] and on a chalcogenide chip [12], extended to multiple wavelength channels without crosstalk [13], to phonon refreshing [14], to storage over 100 pulse widths [15], to cryogenic fiber [16], to phase-encoded data [17], and to polarization multiplexing [18]. Quantum-regime analyses now predict optoacoustic entanglement [19] and a solid-state Brillouin quantum memory [20]. In parallel, SBS has been observed in TFLN waveguides [21–24], and gigahertz phononic waveguides with sub-dB/mm loss at 4 K have been reported in lithium niobate on sapphire [25].

This paper evaluates whether these ingredients can be combined into a *phononic frequency-bin buffer* (PFB) in TFLN. In this all-optical scheme an EO comb supplies the write and read pumps, and no interdigital transducer

(IDT) is needed. The original premise of the concept was that the comb-line amplitudes and phases choose *which* frequency superposition is written into the phonon, and that a later read pulse with an arbitrary comb combination releases it on demand. This would provide time–frequency permutation and photon synchronization. We quantify the delay figures of merit, pump power, and thermal-noise-limited fidelity. We then show that the premise of superposition selection does not survive the phase-matching geometry of backward SBS, and we identify the reduced functionality that remains.

All numbers in this work are simulations or analytic estimates built on literature parameters; no device was fabricated or measured. Parameters and their status are collected in Table I.

II. CONCEPT AND MODEL

A. Geometry and coupled-mode equations

Signal bins $\omega_n = \omega_0 + n\Delta$ ($n = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$; $\Delta/2\pi = 10$ GHz assumed) propagate in $+z$ with group velocity $v_o = c/n_g$. A pump comb with lines $\omega_m^p = \omega_m - \Omega_B$, normalized complex weights w_m ($\sum_m |w_m|^2 = 1$), and common envelope $p(z + v_o t)$ propagates in $-z$. Pair (n, m) beats at frequency $\Omega_B + (n - m)\Delta$ and wave vector $q_{nm} = k_n + |k_m^p| \simeq q_0 + (n + m)\Delta/v_o$. On nanosecond time scales the phonon envelope $B(z, t)$ is effectively local, because it moves only $v_{ac}t \sim 3 \mu\text{m}$ in 1 ns. In a frame

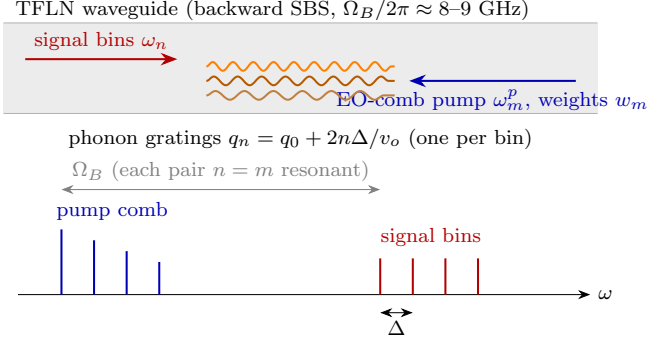


FIG. 1. Concept of the comb-pumped backward-SBS frequency-bin buffer. Signal bins $\omega_n = \omega_0 + n\Delta$ meet a counter-propagating EO-comb pump with lines $\omega_m^p = \omega_m - \Omega_B$ and complex weights w_m . Only pairs $n = m$ drive the phonon resonantly. Because backward SBS requires $q \simeq k_s + |k_p|$, pair n writes a grating with wave vector $q_0 + 2n\Delta/v_o$ (Sec. VI B).

rotating at (Ω_B, q_0) and (ω_n, k_n) , the envelopes obey

$$(\partial_t + v_o \partial_z) A_n = -i g p \sum_m w_m e^{-i\Phi_{nm}} B, \quad (1)$$

$$\partial_t B = -i g p \sum_{n,m} w_m^* e^{i\Phi_{nm}} A_n - \frac{\Gamma}{2} B, \quad (2)$$

$$\Phi_{nm} = \kappa(n+m) \frac{\Delta}{v_o} z - (n-m)\Delta t, \quad (3)$$

where Γ is the phonon energy decay rate. The physical backward-SBS case is $\kappa = 1$. Setting $\kappa = 0$ removes the spatial phase and gives the *local*, single-acoustic-mode model used in the original concept study [Sec. VI A]. The source scripts also include the small Brillouin-frequency dispersion $\epsilon = 2n_g v_{ac}/c = 4.76 \times 10^{-5}$, which detunes pair n by $n\epsilon\Delta$ ($2\pi \times 0.48$ MHz per 10-GHz bin).

B. Pump power for a full swap

Using the standard result that $G_B \Gamma$ is independent of temperature, we take the coupling rate as $g^2 = G_B \Gamma v_o P/4$. The counter-propagating overlap time is $\tau_w/2$. A full photon $\rightarrow$ phonon swap ($g\tau_w/2 = \pi/2$) then requires a peak pump power

$$P_\pi = \frac{4\pi^2}{G_B \Gamma v_o \tau_w^2}, \quad (4)$$

which we evaluate with $\tau_w = 1/B$ for photon bandwidth B . Equation (4) assumes that the signal and pump collide completely inside the waveguide. As discussed in Sec. IV, this requires an optical interaction length of order $v_o \tau$.

TABLE I. Model inputs. “Meas.” denotes a measured literature value. “Assum.” denotes an assumption of this work.

Quantity	Value	Source / status
Brillouin gain G_B	$26.1 \text{ W}^{-1}\text{m}^{-1}$	Meas. (x-cut TFLN, 8.70 GHz) [21]
Brillouin linewidth (RT)	11.9 MHz	Meas. [21]
Brillouin shift used	8.0 GHz	Assum. (device: 8.70 GHz)
Optical group index n_g	2.3	Assum. (not sourced)
TFLN optical loss, record	1.3 dB/m	From $Q_i = 2.9 \times 10^7$ [9]
TFLN optical loss, material	0.2 dB/m	Meas. [10]
TFLN long delay line	0.025 dB/cm, 30 cm	Meas. (abstract) [8]
Fiber loss / group index	0.2 dB/km / 1.468	Assum. (standard SMF)
Acoustic loss RT / 4 K	4.0 / 0.7 dB/mm	Meas. at ≈ 3.4 GHz [25]
Acoustic group velocity	3.1 km/s	Meas. [25]
Acoustic loss at 20 mK	= 4-K value	Assum. (unmeasured)
Bin spacing $\Delta/2\pi$	10 GHz	Assum.

C. Noise-limited fidelity

We assume ideal π -swaps. The write pulse then maps the (vacuum-plus-signal) optical mode onto the phonon and resets the phonon’s thermal occupation, which then rethermalizes. The read pulse retrieves the stored excitation with efficiency $\eta = e^{-\Gamma(t+\tau_w)}$, together with $n_{\text{add}} = n_{th}(1 - e^{-\Gamma(t+\tau_w)})$ thermal phonons, where n_{th} is the Bose occupation at 8 GHz. Each added noise quantum gives a random qubit outcome. Conditioned on a detection, the qubit fidelity is

$$F = \frac{\eta + n_{\text{add}}/2}{\eta + n_{\text{add}}}. \quad (5)$$

Multi-photon events, pump heating, and spontaneous Stokes/anti-Stokes scattering of the pump are not included (Sec. VIII). We use $\Omega_B/2\pi = 8.0$ GHz in n_{th} . The TFLN device that supplies G_B operates at 8.70 GHz [21], where $n_{th}(4 \text{ K}) = 9.1$ instead of 9.9, so our 4-K noise estimate is slightly pessimistic.

III. DELAY FIGURES OF MERIT

Figure 2 and Table II compare loss and physical length per microsecond of delay. Optical delay costs αv_o , acoustic delay costs $\alpha_{ac} v_{ac}$, and a phonon limited by its Brillouin linewidth costs $(10/\ln 10)\Gamma$. With the 4-K acoustic loss of Ref. [25], the phononic delay costs 2.17 dB/ μs . This is a factor 78 below a TFLN optical line at the

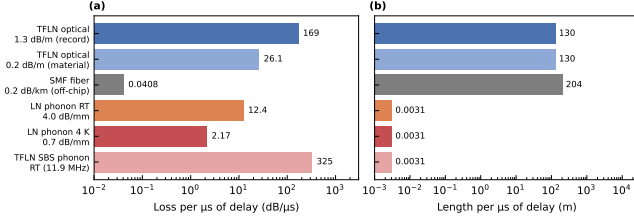


FIG. 2. (a) Loss and (b) physical length per microsecond of delay. Optical rows use $n_g = 2.3$ (TFLN) or 1.468 (fiber). Acoustic rows use the 3.4-GHz loss data of Ref. [25] with $v_{ac} = 3.1$ km/s. The last row converts the measured RT Brillouin linewidth of TFLN [21]. Data: `fig1_delay_line_comparison.csv`.

record 1.3 dB/m (169 dB/ μ s) and a factor 12 below the 0.2-dB/m material limit (26 dB/ μ s). The demonstrated 30-cm TFLN delay line (0.025 dB/cm) [8] corresponds to 326 dB/ μ s. The acoustic wave travels 3.1 mm in 1 μ s, whereas light in TFLN needs 130 m.

Single-mode fiber is better still. At 0.2 dB/km it costs 0.041 dB/ μ s, a factor 53 below the 4-K phonon, but it requires 204 m of fiber per microsecond and chip-to-fiber coupling. The phononic route therefore offers compactness and electrically free, on-chip programmability, not lower loss than fiber. At room temperature the measured SBS linewidth of TFLN (11.9 MHz, ~ 13 ns phonon lifetime) corresponds to 325 dB/ μ s, which is worse than a TFLN optical line. The 4-K figure relies on a different, lower-frequency acoustic mode (see Sec. VIII).

IV. PUMP POWER, BANDWIDTH, AND INTERACTION LENGTH

Equation (4) gives $P_\pi = 15.5$ mW for $B = 10$ MHz, 1.55 W for 100 MHz, and 155 W for 1 GHz [Fig. 3(a)]. The modest gain of TFLN ($G_B = 26.1$ W $^{-1}$ m $^{-1}$) means that only narrowband photons, of order 10–100 MHz, can be swapped at sub-watt to watt peak powers. Photons from cavity-enhanced TFLN-compatible sources are in this range (for example 202 MHz in Ref. [5]), but at 202 MHz the required power already exceeds 6 W. The comparison curve with $G_B = 750$ W $^{-1}$ m $^{-1}$, the value used in the chalcogenide quantum-memory analysis of Ref. [20] where 1.7–3.8 W pulses are considered, assumes the same linewidth and is only indicative.

A constraint that the original concept did not consider sets the device footprint. For the write to swap the whole photon, the signal pulse (τ_s) and the pump pulse (τ_w) must pass through each other completely inside the waveguide. The collision region has length $L_c = v_o(\tau_s + \tau_w)/2$, and it is never shorter than $v_o\tau_s/2$ [Fig. 3(b)]. The written phonon grating spans the same length; its spatial profile is the signal envelope compressed by a factor of two, cf. Ref. [26]. For $\tau_s = \tau_w = \tau$ this gives $L_c = v_o\tau = 13$ m at 10 MHz and 1.3 m

TABLE II. Key figures of merit (simulation/analytic).

Quantity	Value
Phonon delay loss, 4 K (0.7 dB/mm)	2.17 dB/ μ s
Phonon delay loss, RT (4.0 dB/mm)	12.4 dB/ μ s
TFLN SBS phonon, RT (11.9 MHz)	325 dB/ μ s
TFLN optical, 1.3 / 0.2 dB/m	169 / 26.1 dB/ μ s
SMF, 0.2 dB/km	0.041 dB/ μ s
Length per μ s: acoustic / TFLN / SMF	3.1 mm / 130 m / 204 m
P_π at 10 / 100 / 1000 MHz	15.5 mW / 1.55 W / 155 W
Interaction length $v_o\tau$ at 10 / 100 MHz	13 m / 1.3 m
4 K: $\Gamma/2\pi$, n_{th}	79.5 kHz, 9.9
4 K: $\eta(1 \mu$ s), half-efficiency time	0.60, 1.38 μ s
4 K: delay-bandwidth product ($\tau_w = 10$ ns)	138
4 K: $F > 0.9$ / $F > 2/3$ up to 20 mK (assumed loss): $F > 0.99$ up to	40 ns / 0.36 μ s $\geq 10 \mu$ s
Local-model selectivity, $N = 8$, $\tau_w = 1$ ns	0.9988
Space-time selectivity, $N = 2$, backward SBS	0.500

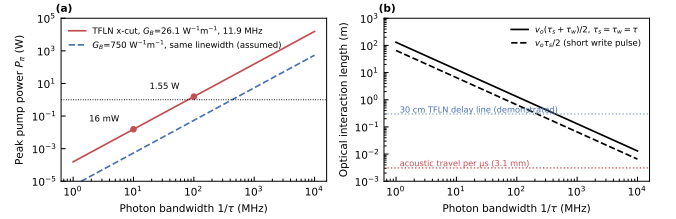


FIG. 3. (a) Peak pump power for a full swap, Eq. (4), versus photon bandwidth $1/\tau$. Points mark 10 and 100 MHz. (b) Optical length of the signal-pump collision region. The acoustic travel per microsecond (3.1 mm) is shown for comparison. Data: `fig2_pump_power.csv`, `derived_numbers.json`.

at 100 MHz. Thus the 3.1 mm per microsecond in Sec. III is the *additional* acoustic travel during storage. The total device must also contain a meter-scale low-loss optical-acoustic waveguide, which would be a spiral far longer than the 30-cm TFLN delay line demonstrated to date [8]. A waveguide shorter than L_c truncates the overlap and raises P_π above Eq. (4). We did not quantify this.

V. STORAGE EFFICIENCY AND NOISE

Figure 4 evaluates Eq. (5) with $\tau_w = 10$ ns. At room temperature the SBS phonon has $n_{th} \approx 768$ and $\Gamma/2\pi = 11.9$ MHz. The fidelity sits at 0.50 for all storage

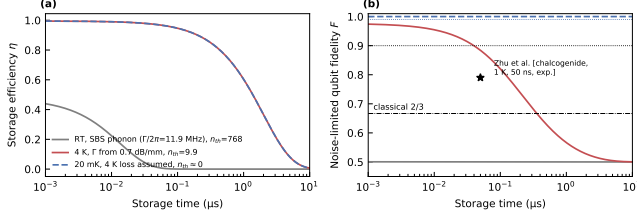


FIG. 4. (a) Storage efficiency and (b) noise-limited qubit fidelity, Eq. (5), with ideal swaps and $\tau_w = 10$ ns. The 20-mK curve assumes the 4-K acoustic loss (unmeasured). The star marks the value reported in Ref. [20] (chalcogenide, 1 K, 50 ns) and is shown only as a reference point. Data: `fig3_storage.csv`.

times, so single photons cannot be stored. At 4 K, the 0.7-dB/mm loss gives $\Gamma/2\pi = 79.5$ kHz ($1/\Gamma = 2.0$ μs), $\eta(1$ $\mu\text{s}) = 0.60$, a half-efficiency time of 1.38 μs , and a delay-bandwidth product of 138. The residual $n_{th} = 9.9$, however, limits the fidelity to 0.976 for $t \rightarrow 0$, to $F > 0.9$ for storage up to 40 ns, and to $F > 2/3$ up to 0.36 μs . For comparison, Ref. [20] reports a fidelity of 0.79 at 1 K and 50 ns for its chalcogenide parameters. At 20 mK ($n_{th} \approx 5 \times 10^{-9}$) the fidelity exceeds 0.99 over the whole computed range up to 10 μs . This result rests on the *assumption* that the 4-K acoustic loss carries over to millikelvin temperatures, which has not been measured for TFLN. In this regime the efficiency, not the fidelity, is the binding constraint: η falls to 0.60 at 1 μs and to 7×10^{-3} at 10 μs .

VI. MULTIMODE OPERATION

A. Local single-acoustic-mode model

In the local limit ($\kappa = 0$) only the resonant pairs $n = m$ survive time averaging over τ_w . Equation (2) then couples the phonon to the single collective mode $a_w = \sum_n w_n^* A_n$, so the comb weights select which superposition is stored. Integrating Eqs. (1)–(2) for one phonon mode and N bins, including the off-resonant $n \neq m$ terms and the $n\epsilon\Delta$ dispersion, with discrete-Fourier-transform (DFT) target modes, we obtain the storage matrix E_{kj} and the selectivity $\eta_k / \sum_j E_{kj}$ (Fig. 5). The selectivity exceeds 0.998 for $\Delta\tau_w/2\pi \geq 8$. For $N = 8$ and $\tau_w = 1$ ns at 10 GHz spacing, both the mean storage efficiency and the mean selectivity are 0.9988. For large $\Delta\tau_w$ the selectivity degrades again (0.9967 at $\Delta\tau_w/2\pi = 128$, $N = 8$), consistent with the accumulated $n\epsilon\Delta\tau_w$ phase. Note also that $\tau_w = 1$ ns corresponds to $P_\pi = 155$ W [Eq. (4)]. At the watt-level operating point ($\tau_w = 10$ ns) the local selectivity lies between the $\Delta\tau_w/2\pi = 64$ and 128 values (0.9990 and 0.9967).

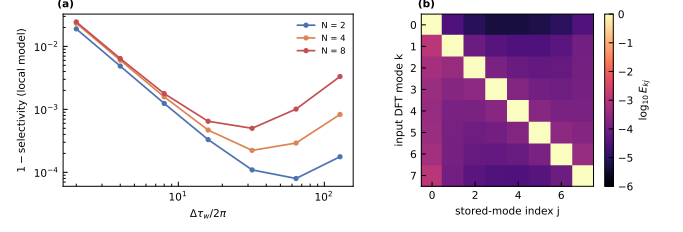


FIG. 5. Local (single-acoustic-mode, $\kappa = 0$) model. (a) 1-selectivity versus $\Delta\tau_w/2\pi$ for $N = 2, 4, 8$ bins. (b) Storage matrix E_{kj} for $N = 8$, $\tau_w = 1$ ns, $\Delta/2\pi = 10$ GHz. Section VIB shows that this model does *not* describe backward SBS. Data: `fig4_multimode.csv`, `fig4b_N8_matrix.csv`.

B. Backward-SBS phase matching invalidates the local model

For $\kappa = 1$, the resonant pair n drives the phonon with spatial phase $e^{i2n\Delta z/v_o}$. Different bins therefore write gratings separated by

$$\Delta q = \frac{2\Delta n_g}{c} \approx 964 \text{ rad/m} \quad (\Delta/2\pi = 10 \text{ GHz}), \quad (6)$$

a spatial beat period of 6.5 mm. The relevant length is the collision region L_c of Sec. IV, which is also the length of the written grating. It is not the acoustic wavelength scale $v_{ac}\tau$ (~ 3 μm). The original concept used the latter length and obtained $\Delta q \ell \approx 3 \times 10^{-3}$, but the correct product is $\Delta q L_c \simeq 2\Delta\tau$. This is ≈ 126 for $\tau = 1$ ns and $\approx 1.3 \times 10^3$ for 10 ns. Because the local model itself requires $\Delta\tau_w \gg 1$ for frequency resolution, resolved bins in backward SBS always write mutually orthogonal gratings. The phonon field is then $B(z) = \sum_n w_n^* e^{i2n\Delta z/v_o} b_n(z)$, with one independent amplitude per bin. This is exactly the mechanism behind the crosstalk-free multi-wavelength storage of Ref. [13].

To check this numerically we integrated Eqs. (1)–(2) on a space-time grid. The signal is advected exactly, the pump is undepleted, $\Gamma \rightarrow 0$, and only the resonant ($n = m$) beats are kept. We used $N = 2$, $\tau_s = 3$ ns, $\tau_w = 1$ ns, $w = (1, 1)/\sqrt{2}$, and a pump area that gives $\eta = 0.93$ in the local model. Figure 6 scales the spatial phase by κ . At $\kappa = 0$ the target mode $(a_0 + a_1)/\sqrt{2}$ is stored with $\eta = 0.926$ and the orthogonal mode is not stored. Including the off-resonant terms gives 1.1×10^{-4} , reproducing the local result. The selectivity falls below 0.99 already at $\Delta q L_c \approx 0.08$. At the physical value $\Delta q L_c = 251$ ($\kappa = 1$) the target and orthogonal modes are stored equally ($\eta = 0.821$ each) and the selectivity is 0.500. A single pump line ($w = (1, 0)$) stores bin 0 with $\eta = 0.926$ and leaves bin 1 untouched ($< 10^{-7}$), confirming bin-selective, not superposition-selective, writing.

The read step is equally constrained. A grating q_{nn} read by pump line m' emits at $\omega_{m'}^p + \Omega_B$. Phase matching requires $k_{\text{out}} + |k_{m'}^p| = q_{nn}$, and energy and momentum conservation together force $m' = n$, up to corrections of order ϵ . Backward SBS therefore returns each stored

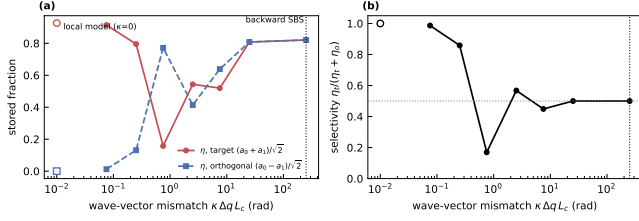


FIG. 6. Space-time check of the mode-selective write ($N = 2$, $\tau_s = 3$ ns, $\tau_w = 1$ ns, $\Delta/2\pi = 10$ GHz). The spatial phase in Eq. (3) is scaled by κ ; the horizontal axis is $\kappa \Delta q L_c$ with $L_c = v_o(\tau_s + \tau_w)/2 = 0.26$ m. Open symbols: $\kappa = 0$ (local model). Dotted line: physical backward SBS ($\kappa = 1$). (a) Stored fraction for the targeted and orthogonal two-bin modes. (b) Selectivity. Data: `data_spatial_scan.json`.

component at its original frequency. It cannot remap frequencies, and a read with an “arbitrary comb combination” only chooses *which* bins are released, with which amplitude and phase.

C. What survives: a bin-diagonal programmable buffer

In backward SBS the comb therefore acts diagonally in frequency. Each pump line sets, for its own bin, the swap angle (via $|w_n|$), a phase (via $\arg w_n$), and the write and read times. This supports two functions:

- *Per-bin programmable delay*, i.e. time reordering of frequency components. This is a permutation in time, not in frequency.
- *Synchronization* of photons that arrive in different bins at different times, by firing each bin’s line at that photon’s arrival and all lines together at read-out.

Frequency-bin superpositions are stored coherently, as N independent gratings. Each fully swapped bin needs its own line at P_π , so simultaneous operation of N bins needs NP_π in total, for example 12.4 W for $N = 8$ at 100 MHz. Mode-selective storage or frequency remapping would require all pairs to address the *same* acoustic mode, i.e. $\Delta q L_c \lesssim 0.1$ (Fig. 6). Forward intra- or inter-modal Brillouin scattering has $\Delta q \approx 0$ or an engineered Δq . A forward-Brillouin “super- π ” pulse that swaps one phonon with an arbitrary frequency-bin superposition, and performs quantum frequency translation through the phonon, has already been proposed [27]. A TFLN implementation of that geometry is not evaluated here.

VII. RELATION TO PRIOR WORK AND NOVELTY

Table III positions the reduced PFB among related approaches. Classical Brillouin storage in chalcogenide

chips [12, 14, 15] and in fiber [11, 16–18] established coherent storage. Multi-wavelength, crosstalk-free operation [13] already demonstrated the bin-diagonal function that survives our analysis. The quantum-regime analysis of Ref. [20] treats a single optical mode. Superposition-selective swaps and frequency translation via a phonon have been proposed for forward Brillouin [27]. Acousto-optic intermodal frequency beam splitters (FRODO) give universal frequency-bin unitaries but no storage [28]. Outside Brillouin physics, a Raman memory in warm vapor already performs mode-selective storage, filtering and conversion of 30 temporal modes [29]. Spectrally multiplexed atomic-frequency-comb memories [30] and a recent proposal for frequency-bin synthesis in a rare-earth memory [31] address frequency bins with different physics. TFLN SBS has been characterized for gain, lasing, and RF photonics [21–24], and IDT-driven backward-Brillouin coupling has been demonstrated in LN-on-sapphire rings [32, 33]. We found no report of Brillouin optical *storage* in TFLN, or of an EO-comb-pumped Brillouin frequency-bin memory. The contributions of this work are therefore limited to (i) TFLN-specific delay, power and noise budgets; (ii) the explicit space-time demonstration that comb-programmed superposition selection fails in backward SBS; and (iii) the interaction-length constraint on footprint. We regard the novelty as moderate, and lower than originally envisaged.

VIII. LIMITATIONS

The conclusions above are subject to the following limitations, which we consider essential:

1. *The acoustic loss of TFLN at millikelvin temperatures is unmeasured.* All 20-mK results assume the 4-K value of 0.7 dB/mm.
2. *Frequency mismatch of the loss data.* The 0.7-dB/mm (4 K) and 4.0-dB/mm (RT) values were measured at ≈ 3.4 GHz in LN-on-sapphire phononic waveguides [25]. The backward-SBS phonon is at ≈ 8 –9 GHz in a different (SAW-like) mode of a TFLN optical waveguide [21]. Acoustic loss generally increases with frequency, so the 8-GHz loss may be substantially larger than assumed.
3. *Not computed:* pump-induced heating (which would raise n_{th} above the bath value, critically at 20 mK with watt-level pumps), spontaneous Stokes and anti-Stokes noise generated by the pump, and chip coupling loss. Off-resonant $n \neq m$ crosstalk is included only in the local model.
4. The meter-scale collision region (Sec. IV) requires long, low-loss waveguides that guide the 8-GHz phonon throughout. Such waveguides have not been demonstrated, and their optical loss (e.g. ≈ 1.3 –2.5 dB/m) adds a fixed insertion loss that is not included in η .

TABLE III. Comparison with related frequency/temporal-mode processors and memories. Values are as reported in the cited works. “This work” rows are simulations.

Approach	Platform	Storage	Mode/bin operation	Reported metrics
PFB, local model (this work)	TFLN, backward SBS	yes (sim.)	superposition-selective (invalid, Sec. VIB)	sel. 0.9988 ($N = 8$)
PFB, space-time (this work)	TFLN, backward SBS	yes (sim.)	bin-diagonal: per-bin delay/phase	sel. 0.50 ($N = 2$); 2.2 dB/ μ s (4 K)
EO QFP [3]	fiber EOM + shaper	no	arbitrary unitaries	Hadamard $F = 0.99998$, $P = 0.9739$
Double-ring f-BS [5]	TFLN, EO	no	2-bin BS, CZ	1-qubit 97.1%, CZ 91.4%
QPG [6]	TFLN, SFG	no	temporal-mode selection	sel. 96.8% / 91.7% (1 photon)
FRDO [28]	Si/SiN, intermodal AO	no	universal unitaries (theory)	7-GHz bins
Forward super- π [27]	forward Brillouin	phonon	superposition swap, QFT (theory)	—
Raman TM processor [29]	warm Cs vapor	yes	30 HG modes, conversion	5D process $F = 0.943$
Brillouin quantum memory [20]	chalcogenide (theory)	yes	single mode	$F = 0.79$ (1 K, 50 ns)

5. $n_g = 2.3$, $\Delta/2\pi = 10$ GHz, ideal π -swaps, undepleted pumps, and the absence of acoustic diffraction and dispersion are assumptions. The space-time check uses $N = 2$ and $\Gamma \rightarrow 0$ and does not optimize pulse shapes.

IX. CONCLUSION

A comb-pumped backward-SBS buffer in TFLN offers a compact, IDT-free acoustic delay that, at 4 K, is an order of magnitude less lossy per microsecond than the best TFLN optical lines but still worse than fiber. Its pump power restricts it to ~ 10 –100-MHz photons, and its noise restricts single-photon use to cryogenic, and realistically millikelvin, operation. Programmable selection of frequency superpositions, the central premise of

the concept, is incompatible with backward-SBS phase matching. The device is instead a bin-diagonal memory with per-bin delay, phase and swap control, already anticipated classically by multi-wavelength Brillouin storage. Mode-selective storage would require a forward-Brillouin geometry, which we leave to future work. Measurements of 8-GHz acoustic loss at millikelvin temperatures, and of pump-induced heating, are the decisive experimental inputs.

DATA AND CODE AVAILABILITY

All scripts and data are provided with the manuscript: `scripts/simC.py` and `params.py` (original model), `paper/scripts/make_figs.py` (figures and derived numbers), and `paper/scripts/spatial_check.py` and `spatial_scan.py` (space-time check).

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